Quant Finance Projects

Concepts & What to Know

Four projects · Core theory · Interview-ready explanations

SVR · Portfolio Optimization · Monte Carlo · Yield Curve

PROJECT 1

Support Vector Regression

Predicting equity returns with machine learning

What the project does

Trains an ε-SVR model on engineered features (lagged returns, moving averages, RSI) to predict next-day AAPL returns. Uses walk-forward cross-validation to avoid lookahead bias and builds a long-biased overlay strategy from the model's signal.

Core concepts you must understand

1. Support Vector Regression (SVR)

SVR finds a function f(x) that fits most training points within an ε-tube around the prediction, ignoring errors smaller than ε. Only the points outside the tube — called **support vectors** — determine the solution. This makes SVR robust to outliers and well-suited for sparse financial signals.

```
minimize: ½||w||² + C·Σ(ξ■ + ξ■*)
```

```
subject to: y■ - f(x■) ≤ ε + ξ■ and f(x■) - y■ ≤ ε + ξ■*
```

- **C** — regularization: high C = fits training data closely, risks overfitting
- ε — tube width: predictions within ε of truth incur zero loss

2. The Kernel Trick

The RBF kernel implicitly maps inputs into a high-dimensional space where a linear separator exists, without ever computing the high-dimensional coordinates explicitly.

```
K(x, x') = exp(-γ ||x - x'||²)
```

γ controls how fast similarity decays with distance. High γ = narrow Gaussians = model only responds to very nearby points.

3. The ε Calibration Problem

This is the most important technical insight in the project. If ε is too large relative to std(y), all training points fall inside the insensitive tube — the model sees no error to minimize, finds no support vectors, and predicts the mean (≈ 0) for everything.

```
Rule of thumb: ε should be ≈ 1–5% of std(returns)
```

std(AAPL daily returns) ≈ 0.013. Original ε = 0.1 → ε/σ ≈ 7.7 → flat predictions.

4. Directional Accuracy vs MSE

In an efficient market, MSE-optimal predictions converge to near-zero (the unconditional mean is the best forecast). A model optimized on MSE produces flat predictions that are useless for trading. Directional accuracy — did the model get the sign right? — is the correct scoring metric for a trading signal.

5. Walk-Forward Validation

Standard k-fold cross-validation leaks future data into training. TimeSeriesSplit enforces that training data always precedes the test fold in time. This is non-negotiable for financial ML.

6. Long-Biased Overlay Strategy

Instead of a pure long/short strategy (unfair benchmark against a 500%+ bull market), the model tilts position size: always long, but 30%–150% exposure depending on SVR confidence. This frames the model as a signal for sizing, not a direction-picker.

Why Vasicek fits worse — aside

> ***This is addressed in the Yield Curve section. Skip ahead if reviewing that project.***

Key numbers to remember

- Best params: C=50, ε=0.0009 (≈ 7% of std), γ=RBF default
- Directional accuracy: ~52–54% — small but consistent edge
- Epsilon must be calibrated relative to target standard deviation, not set arbitrarily

PROJECT 2

Portfolio Optimization

Markowitz, risk parity, CVaR, and Black-Litterman

What the project does

Implements five portfolio construction methods on a multi-asset universe (SPY, TLT, GLD, EFA, IEF), runs a rolling walk-forward backtest with transaction costs, tests whether Sharpe ratio differences are statistically significant, and adds regime analysis and rolling correlations.

Core concepts you must understand

1. Minimum Variance (Markowitz, 1952)

Finds the portfolio with the lowest variance for a given set of assets. This is a quadratic program:

```
min w'Σw s.t. 1'w = 1, w ≥ 0
```

The solution concentrates in low-volatility, low-correlation assets. It ignores expected returns entirely — useful when return forecasts are unreliable.

2. Maximum Sharpe Ratio

Maximizing $(\mu'w - rf) / \sqrt{w'\Sigma w}$ is a fractional program. The standard trick substitutes $y = w / \sqrt{w'\Sigma w}$:

```
max (μ - rf)'y s.t. y'Σy ≤ 1, y ≥ 0
```

```
then normalize: w = y / sum(y)
```

This converts a non-convex ratio into a convex QP — solvable with standard solvers (CVXPY/CLARABEL).

3. Risk Parity / Equal Risk Contribution

Each asset contributes equally to total portfolio variance:

```
w■ · (Σw)■ = w'Σw / n for all i
```

No closed form exists. Solved via cyclical coordinate descent: update one weight at a time holding others fixed, using the positive root of a quadratic equation.

4. CVaR Optimization (Rockafellar-Uryasev, 2000)

CVaR (Expected Shortfall) is the expected loss in the worst α% of scenarios. Minimizing it is an LP:

```
min ζ + (1/Tα) · Σ u■
```

```
s.t. u■ ≥ -r■'w - ζ, u■ ≥ 0, 1'w = 1
```

CVaR is a coherent risk measure (unlike VaR). The LP formulation is exact, not an approximation.

5. Ledoit-Wolf Shrinkage Covariance

The sample covariance matrix is noisy with few observations relative to assets. Ledoit-Wolf shrinks it toward a structured target:

```
Σ■ = (1-α)·S + α·μ·I
```

α is chosen analytically to minimize mean squared error. This reduces estimation error and prevents near-singular matrices.

6. Black-Litterman Model

Combines market equilibrium returns with investor views using Bayes' theorem.

```
Π = δΣw_mkt (implied equilibrium returns)
```

```
μ_BL = [(τΣ)■¹ + P'Ω■¹P]■¹ [(τΣ)■¹Π + P'Ω■¹Q]
```

- **P** — pick matrix encoding views (relative or absolute)
- **Q** — view return vector
- Ω — view uncertainty (proportional to $\tau\Sigma$ by default)
- τ — scalar controlling prior uncertainty (typically 0.01–0.10)

7. Jobson-Korkie Test for Equal Sharpe Ratios

Most projects just compare Sharpe ratios numerically. The JK test asks: is the difference statistically significant?

```
z = (μ■σ■ - μ■σ■) / √(θ/T)
```

θ accounts for the correlation between strategies and the moments of their return distributions. $p < 0.05$ means the Sharpe difference is unlikely to be noise.

8. Turnover and Transaction Costs

One-way turnover = Σ|Δw■| / 2. At 10 bps cost per unit turnover, a strategy that turns over 50% of the portfolio monthly pays 0.10% × 0.5 × 12 = 0.60% per year in friction — enough to erode a moderate Sharpe ratio.

Key numbers to remember

- Rebalance quarterly (63 days), 2-year estimation window

- Transaction cost: 10 bps one-way per unit turnover
- Per-asset weight cap: 40%
- JK test: $p < 0.05$ = significant Sharpe difference

PROJECT 3

Monte Carlo Option Pricing

GBM simulation, Black-Scholes validation, antithetic variates

What the project does

Prices European call options by simulating terminal stock prices under risk-neutral GBM. Benchmarks against Black-Scholes, studies convergence as N increases, applies antithetic variates for variance reduction, and calibrates volatility from historical SPY returns.

Core concepts you must understand

1. Geometric Brownian Motion (GBM)

Stock prices follow GBM under the physical measure:

```
dS■ = μS■dt + σS■dW■
```

For option pricing we switch to the risk-neutral measure, replacing μ with r:

```
dS■ = rS■dt + σS■dW■
```

This is why the simulation uses r as drift, not the expected return μ. Under the risk-neutral measure, all assets grow at the risk-free rate on average — this is the no-arbitrage condition.

2. Euler-Maruyama Discretization (log-space)

To avoid negative prices and reduce discretization error, we simulate in log-space:

```
S_{t+Δt} = S■ · exp[(r - ½σ²)Δt + σ√Δt · Z] , Z ~ N(0,1)
```

The $(r - \frac{1}{2}\sigma^2)$ adjustment is Itô's correction — it appears because log(S) and S have different drift terms when applying Itô's lemma.

3. Risk-Neutral Pricing

The option price is the discounted expected payoff under Q (the risk-neutral measure):

```
C = e^{-rT} · E^Q[max(S_T - K, 0)]
```

Monte Carlo estimates this expectation by averaging discounted payoffs across N simulated paths.

4. Central Limit Theorem and Standard Error

By the CLT, the MC estimator is asymptotically normal:

```
SE = s / √N where s = sample std of discounted payoffs
```

SE halves every time N quadruples. To get one extra decimal place of accuracy costs 100× more simulations. This $1/\sqrt{N}$ convergence rate is the fundamental limitation of Monte Carlo.

5. Antithetic Variates

For each random draw Z, also simulate -Z. The two paths are negatively correlated, so their average has lower variance than either alone:

```
Var[(f(Z) + f(-Z))/2] = ½·Var[f(Z)] + ½·Cov[f(Z), f(-Z)]
```

When f is monotone in Z (as the call payoff is), the covariance is negative and variance is strictly reduced — same simulation budget, tighter confidence intervals.

6. Black-Scholes as Ground Truth

```
C = S■Φ(d■) - Ke^{-rT}Φ(d■)
```

```
d■ = [ln(S■/K) + (r + ½σ²)T] / (σ√T) d■ = d■ - σ√T
```

Black-Scholes is the closed-form limit of MC as $N \to \infty$ under the same GBM assumptions. The MC error should shrink toward zero as N grows — this is the primary validation test.

7. Why MC matters in practice

For vanilla European options, Black-Scholes is always faster and exact — you would never use MC here. MC earns its place for:

- Path-dependent payoffs: Asian (average price), barrier (knock-in/out)
- American options: requires Longstaff-Schwartz least-squares MC
- Multi-asset options: basket options, correlation structures
- Stochastic vol models (Heston): no simple closed form

Key numbers to remember

- At N=10,000: MC price 9.31 vs BS 9.41 — abs error 0.10 (1.1%)
- At N=50,000: abs error shrinks to 0.10 — consistent with $1/\sqrt{N}$ decay
- SE at N=10,000: 0.14 → 95% CI width ≈ ±0.28
- Itô correction: drift in log-space = $r - \frac{1}{2}\sigma^2$, not r

PROJECT 4

Yield Curve Modeling

Bootstrap, Nelson-Siegel, and Vasicek

What the project does

Bootstraps zero (spot) rates from par Treasury yields, fits the Nelson-Siegel parametric model, calibrates a Vasicek short-rate model, and simulates interest rate paths showing mean-reversion to the long-run equilibrium rate.

Core concepts you must understand

1. Par Yield vs Spot Rate vs Forward Rate

- **Par yield** — the coupon rate that makes a bond price at face value. This is what FRED reports.
- **Spot rate r(T)** — the yield on a zero-coupon bond maturing at T. This is what you actually use for discounting.
- **Forward rate f(T)** — the rate implied for a future period [T, T+dt]. Derived from the slope of the spot curve.

```
Discount factor: P(0,T) = e^{-r(T)·T}
```

```
Forward rate: f(T) ≈ d[r(T)·T]/dT
```

2. Bootstrapping

Par bonds price at 100 by construction. For each maturity, the spot rate is the discount rate that makes the bond price equal 100, given that all shorter-maturity spot rates are already known:

```
P = Σ c·e^{-r(t■)·t■} + (c+F)·e^{-r(T)·T} = 100
```

Rearranging for r(T): subtract the PV of known coupon cash flows, then solve the last term. This requires semi-annual coupon bonds — T-bills (≤ 6 months) are zero-coupon and treated separately.

3. Nelson-Siegel Model

```
r(τ) = β■ + β■·(1-e^{-τ/λ})/(τ/λ) + β■·[(1-e^{-τ/λ})/(τ/λ) - e^{-τ/λ}]
```

- β■ — long-run level (r(τ) → β■ as τ → ∞)
- β■ — slope (r(0) - r(∞) = β■ + β■ at the short end)
- β■ — curvature (controls mid-maturity hump)
- λ — where the curvature term peaks (larger λ = hump at longer maturity)

RMSE of 11 bps on this project — tight fit because NS was designed for this curve shape.

4. Why Vasicek Fits Worse Than Nelson-Siegel

This is the key conceptual point. NS is a *curve-fitting model* — three parameters are tuned specifically to minimize fitting error. Vasicek is an *equilibrium model* — parameters have economic meaning (mean-reversion speed, long-run rate, volatility) and the yield curve shape is a by-product.

Vasicek implies yields of the form:

```
y(τ) = θ + (r■ - θ)·B(τ)/τ - σ²/(4κ)·B(τ)²/τ
```

```
where B(τ) = (1 - e^{-κτ}) / κ
```

This is necessarily smooth and monotone-to-flat — it cannot reproduce curves with a pronounced long-end upturn (like 20y > 10y > 30y). NS has no such structural constraint, so it fits better almost by construction.

> ***Put simply: NS is a flexible curve-fitter. Vasicek is an interest rate model that happens to imply a curve. You would use Vasicek for simulation and derivatives pricing, not for exact curve fitting.***

5. Vasicek Model

```
dr■ = κ(θ - r■)dt + σdW■
```

- κ — mean-reversion speed. Half-life = $\ln(2)/\kappa$
- θ — long-run mean. Rates are pulled toward θ over time
- σ — instantaneous volatility of the short rate

Vasicek is analytically tractable: zero-coupon bond prices have a closed form, making calibration fast. Drawback: rates can go negative (unlike CIR which enforces $r \geq 0$).

6. Mean-Reversion and the Half-Life

If the short rate is currently below θ, the drift term $\kappa(\theta - r)$ is positive — the rate is pulled upward. The speed of this pull is κ. The half-life tells you how long it takes for half the gap between r■ and θ to close:

```
half-life = ln(2) / κ
```

Calibrated value: $\kappa = 1.11$, half-life ≈ 0.6 years. This means the market is in a regime where rates are expected to normalize relatively quickly.

7. The Term Structure of Interest Rates

The shape of the yield curve encodes market expectations:

- **Normal (upward-sloping)**: long rates > short rates. Growth and inflation expected.
- **Inverted (downward-sloping)**: short rates > long rates. Often precedes recessions.
- **Flat/humped**: transition between regimes.

The 2023 US curve this project models is inverted at the short end, flattening toward a 4.4% long-run level — consistent with the Fed hiking cycle and market pricing of eventual cuts.

Key numbers to remember

- Nelson-Siegel RMSE: 0.11% (11 bps) — standard for academic fitting
- Vasicek RMSE: 0.20% — higher because it is structurally constrained
- Vasicek θ = 4.38%, half-life ≈ 0.6 years
- Forward rate > spot rate when curve is upward sloping (and vice versa)

Quick Reference — Formulas to Know

Risk and Performance

```
Sharpe = (E[r] - rf) / σ
```

```
Sortino = (E[r] - rf) / σ_downside
```

```
CVaR(α) = E[loss | loss > VaR(α)]
```

```
Max Drawdown = min(S■/max(S■, s≤t) - 1)
```

Options

```
C = S■Φ(d■) - Ke^{-rT}Φ(d■)
```

```
d■ = [ln(S/K) + (r+½σ²)T] / σ√T d■ = d■ - σ√T
```

```
Put-Call Parity: C - P = S■ - Ke^{-rT}
```

Portfolio Theory

```
Min-Var: min w'Σw s.t. 1'w=1
```

```
Max-Sharpe: max (μ-rf)'y s.t. y'Σy≤1 → normalize
```

```
Risk Parity: w■(Σw)■ = w'Σw/n for all i
```

```
BL posterior: μ_BL = [(τΣ)■¹+P'Ω■¹P]■¹[(τΣ)■¹Π+P'Ω■¹Q]
```

Fixed Income

```
Spot → Discount: P(0,T) = e^{-r(T)·T}
```

```
Vasicek: dr■ = κ(θ-r■)dt + σdW■
```

```
NS: r(τ) = β■ + β■·L(τ) + β■·C(τ)
```

Statistics

```
MC std error: SE = s/√N
```

```
JK test: z = (μ■σ■ - μ■σ■)/√(θ/T)
```

```
Bootstrap CI: [percentile(α/2), percentile(1-α/2)]
```